Agnostic learning of circuit classes under randomized average-case easiness of NP

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doi:10.4230/LIPIcs.CCC.2022.16; CC BY 4.0

Editorial scope. Counted original Theorem56/restated63 once, the main conditional agnostic learner for circuits under time/advice-bounded samplable distributions. Original randomized average-case NP easiness assumption, parameter ranges and printed sample/time bounds retained. Complete sampling-depth, probabilistic approximation, refuter, entropy-encoding and incompressibility arguments included. Imported coding and refutation-to-learning theorems retained as exact background statements; support lemmas are not extra tasks.

Allowed standard prerequisites (editorial).

- Imported RRHS-refutation-to-agnostic-learning theorem [21], unconditional pKt coding theorem [34], classical Kolmogorov entropy bound [27] and DPG reconstruction [11]. Exact imported statements retained. All source-local pKt reconstruction/symmetry/approximation, sampling-depth, entropy encoding, refuter and final application proofs included. Imported theorems are not recursively reproved.

2 Preliminaries (necessary author context)

2.1 Notation

We say that a function $p : \mathbb{N} \to \mathbb{N}$ is non-decreasing if $p(\ell) \geq \ell$ for every integer $\ell \geq 0$. Throughout the paper, we will rely on constructions whose associated complexity parameters (e.g., running time) can be bounded by some polynomial p. We will implicitly assume that the polynomials provided in these results are non decreasing in order to avoid including this condition in every statement. This can be done without loss of generality by the monotonicity of our bounds. For a probability distribution D_n, we use $\mathrm{supp}(D_n)$ to denote its support. Given an element x, we let $D_n(x)$ denote the probability of x under D_n. We denote by U_n the uniform distribution over n-bit strings. Let D_1 and D_2 be probability distributions supported over a set X, $A : X \to \{0,1\}$, and $\varepsilon \geq 0$. We say that A ε-distinguishes D_1 and D_2 if

$$\left| \Pr_{x \sim D_1}[A(x) = 1] - \Pr_{x \sim D_2}[A(x) = 1] \right| \geq \varepsilon.$$

If A is a randomized algorithm, each probability in the expression above also takes into account the internal randomness of A, which will be denoted by r_A.

2.2 Average-case complexity

Recall that a pair $(L, \mathcal{D})$ is a distributional problem if $L \subseteq \{0,1\}^*$ and $\mathcal{D} = \{D_n\}_{n\geq 1}$ is an ensemble of probability distributions, where each D_n is supported over $\{0,1\}^*$. Let $\mathcal{D} = \{D_n\}_{n\geq 1}$ be an ensemble of distributions. We say that $\mathcal{D} \in$ PSamp if there is a randomized polytime algorithm A such that, for every $n \geq 1$, $A(1^n)$ is distributed according to D_n. More generally, we use $D_n \in$ Samp$[T(n)]/a(n)$ to denote that D_n can be sampled by an algorithm that runs in time $T(n)$ and has advice complexity $a(n)$. We let DistNP denote the set of distributional problems $(L, \mathcal{D})$ with $L \in$ NP and $\mathcal{D} \in$ PSamp.

Definition 4 (Avg$_\delta$BPP **[3]).** Let $(L, \mathcal{D})$ be a distributional problem, and $\delta : \mathbb{N} \to [0,1]$. We say that $(L, \mathcal{D}) \in$ Avg$_\delta$BPP if there is randomized polytime algorithm A such that

1. for every $n > 0$, and every $x \in \text{supp}(D_n)$,
$$\Pr_{r_A}[A(x;n) \notin \{L(x), \perp\}] \leq 1/4,$$
2. for every $n > 0$,
$$\Pr_{x\sim D_n}\left[\Pr_{r_A}[A(x;n) = \perp] \geq 1/4\right] \leq \delta(n),$$

where r_A denotes the internal randomness of A. Such an algorithm A is called an randomized errorless heuristic for $(L, \mathcal{D})$ with failure probability at most δ.

Definition 5 (AvgBPP [3]). [1] We say that $(L, \mathcal{D})$ is in AvgBPP if there exist a randomized algorithm A and a polynomial p such that

1. for every $n, \delta > 0$, and every $x \in \text{supp}(D_n)$,
$$\Pr_{r_A}[A(x;n,\delta) \notin \{L(x), \perp\}] \leq 1/4,$$
2. for every $n, \delta > 0$,
$$\Pr_{x\sim D_n}\left[\Pr_{r_A}[A(x;n,\delta) = \perp] \geq 1/4\right] \leq \delta(n),$$
3. for every $n, \delta > 0$, and every $x \in \text{supp}(D_n)$, $A(x;n,\delta)$ runs in time at most $p(n/\delta)$.

Such an algorithm A is called a randomized errorless heuristic scheme for $(L, \mathcal{D})$.

Lemma 6. Let $(L, \mathcal{D})$ be a distributional problem in Avg$_{n^{-1}}$BPP. There exists a randomized polynomial-time algorithm B such that

1. for every $n > 0$, and every $x \in \text{supp}(D_n)$, $B(x;n) \in \{0,1\}$,
2. for every $n > 0$, and every $x \in \text{supp}(D_n)$, if $x \in L$, then $\Pr_{r_B}[B(x;n) = 1] \geq 1 - 1/n$, and
3. for every $n > 0$, $\Pr_{x\sim D_n, r_B}[B(x;n) = L(x)] \geq 1 - 3/n$.

[1] Some authors define AvgBPP as $\bigcap_{c>0}$ Avg$_{n^{-c}}$BPP. All our results also hold with respect to that definition.

Proof. Let A be an $\mathrm{Avg}_{n^{-1}}\mathrm{BPP}$ algorithm for $(L, \mathcal{D})$ as in Definition 4. Let A' be the algorithm obtained, using standard amplification techniques,[2] such that for all $x \in \mathrm{supp}(D_n)$,

$$\Pr_{r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \leq 1/n \tag{1}$$

and

$$\Pr_{x \sim D_n}\left[\Pr_{r_{A'}}[A'(x;n) = \perp] \geq 1/n\right] \leq 1/n. \tag{2}$$

Let B be the randomized algorithm which, on input $x \in \mathrm{supp}(D_n)$, simulates $A'(x;n)$ and outputs 1 if $A'(x;n) \in \{1, \perp\}$ or 0 if $A'(x;n) = 0$. By Equation (1), if $x \in L$, then

$$\Pr_{r_B}[B(x;n) = 0] = \Pr_{r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \leq 1/n.$$

Let $\mathrm{bad} := \{x \in \mathrm{supp}(D_n) \mid \Pr_{r_{A'}}[A'(x;n) = \perp] \geq 1/n\}$. By Equation (2), at most a $(1/n)$-measure of $x \in \mathrm{supp}(D_n)$ are in bad. Thus,

$$\begin{aligned}
\Pr_{x \sim D_n, r_B}[B(x;n) \neq L(x)] &\leq \Pr_{x, r_{A'}}[A'(x;n) = \perp] + \Pr_{x, r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \\
&\leq \Pr_{x}[x \in \mathrm{bad}] + \Pr_{x, r_{A'}}[A'(x;n) = \perp \mid x \notin \mathrm{bad}] \\
&\quad + \Pr_{x, r_{A'}}[A'(x;n) \notin \{L(x), \perp\}] \\
&\leq 1/n + 1/n + 1/n = 3/n.
\end{aligned}$$

This completes the proof. ◁

2.4 Direct product generators (DPGs)

Definition 13 (Direct Product Generator (DPG) [11, Definiton 3.10]). For $k, n \in \mathbb{N}$, we define the k-wise direct product generator to be the function

$$\mathrm{DP}_k : \{0,1\}^n \times \{0,1\}^{nk} \to \{0,1\}^{nk+k}$$

such that

$$\mathrm{DP}_k(x; z^1, \ldots, z^n) := (z^1, \ldots, z^k, x \cdot z^1, \ldots, x \cdot z^k).$$

Lemma 14 (DPG Reconstruction [11, Lemma 3.14]). For any $n, k \in \mathbb{N}$ with $k \leq 2n$, and $\varepsilon > 0$, there exists a pair of algorithms A and $\mathrm{Recon}^{(-)}$ such that

- Recon^{D} takes oracle access to an oracle $D : \{0,1\}^{nk+k} \to \{0,1\}$.
- $A : \{0,1\}^n \times \{0,1\}^r \to \{0,1\}^k$ is called the advice function and is computable in time $\mathrm{poly}(n/\varepsilon)$.
- $\mathrm{Recon}^{(-)} : \{0,1\}^k \times \{0,1\}^r \to \{0,1\}^n$ is called a reconstruction procedure and is computable in time $\mathrm{poly}(n/\varepsilon)$.[3]

[2] In more detail, A' can be obtained from A by running it $O(n)$ times and selecting the most common output.

[3] We note that if the oracle is a uniform algorithm that runs in time $t(m)$ on inputs of length m, the reconstruction procedure can be computed in time $t(nk+k) \cdot \mathrm{poly}(n/\varepsilon)$.

- The randomness complexity r is at most $\text{poly}(n/\varepsilon)$.
- For any string $x \in \{0,1\}^n$ and any function D that ε-distinguishes $\text{DP}_k(x; U_{nk})$ from U_{nk+k}, it holds that

$$\Pr_{w \sim \{0,1\}^r}[\text{Recon}^D(A(x,w),w) = x] \geq 1/\text{poly}(n/\varepsilon).$$

2.7 Encoding tuples (author context)

We will also need to consider distributions that generate input instances that are tuples, while the only known information to the distributions is the length of the instances. To do this, we will encode the format as the length of the instances. More specifically, we use the following way to encode tuples [3].

Proposition 16 ([3]; see also [6, Proposition 3.1]). There is a pair of polytime encoding-decoding algorithms $\text{Enc} : \mathbb{N}^* \to \mathbb{N}$ and $\text{Dec} : \mathbb{N} \to \mathbb{N}^* \cup \{\perp\}$ such that

1. For $k \in \mathbb{N}$ and k integers $n_1, n_2, \ldots, n_k$, we write $\langle n_1, n_2, \ldots, n_k \rangle := \text{Enc}(n_1, n_2, \ldots, n_k)$, and we have
   (a) $\sum_{i=1}^{k}(2 \cdot (\lfloor \log n_i \rfloor + 2) + n_i) \leq \langle n_1, \ldots, n_k \rangle \leq O_k(\prod_{i=1}^{k}(n_i \cdot \log^2 n_i))$.
   (b) $\text{Dec}(\langle n_1, n_2, \ldots, n_k \rangle) = n_1, \ldots, n_k$. Here, Dec takes the binary representation of an integer and outputs k integers which are also represented in binary.
2. $\text{Dec}(u) = \perp$ if $u \neq \langle \vec{n} \rangle$ for any $\vec{n} \in \mathbb{N}^*$.

Proof. Given k integers $n_1, n_2, \ldots, n_k$, the encoding algorithm Enc first obtains the binary string

$$z := \alpha_1 \circ 01 \circ \alpha_2 \circ 01 \circ \alpha_k \circ 01 \circ \text{bin}(n_1) \circ \text{bin}(n_2) \circ \cdots \circ \text{bin}(n_k),$$

where for each $i \in [k]$, α_i is obtained from $\text{bin}(|\,\text{bin}(n_i)|)$ with every bit doubled. Then Enc outputs the integer n whose binary representation is z. It is easy to see that given the binary representation of n, which is simply the string z, one can recover the $n_1, n_2, \ldots, n_k$ in time polynomial in $|z|$. Also, item (a) can be verified by a simple calculation. ◁

3 Probabilistic Time-Bounded Kolmogorov Complexity (author excerpts)

In this section, we formally define probabilistic Kolmogorov complexity and prove some useful properties of this notion, which will be used later in proving our main results.

Definition 17 (pK^t). For strings $x, y \in \{0,1\}^*$, a time bound $t \in \mathbb{N}$, an oracle A, and $\delta \in [0,1]$, the δ-probabilistic A-oracle t-bounded Kolmogorov complexity of x given y is defined as

$$\mathrm{pK}^{t,A}_\delta(x \mid y) := \min\left\{k \;\middle|\; \Pr_{w\sim\{0,1\}^t}[\exists M \in \{0,1\}^k, M^A(w,y) \text{ outputs } x \text{ within } t \text{ steps}] \geq \delta\right\},$$

where M is a RAM-machine.[4] We omit the subscript δ if $\delta = 2/3$, omit the superscript A if $A = \emptyset$, omit "$\mid y$" if y is the empty string, and omit the superscript t if $t = \infty$. For simplicity, in the rest of this section we only consider pK^t without oracles. It is easy to see that all the results also hold with any oracle.

Lemma 21 (Success Amplification). For any string $x \in \{0,1\}^n$, time bound $t \in \mathbb{N}$, and $0 \leq \alpha < \beta \leq 1$, we have

$$\mathrm{pK}^{O(qt/\alpha)}_\beta(x) \leq \mathrm{pK}^t_\alpha + O(\log(q/\alpha)),$$

where $q = \ln(1/(1-\beta))$.

Proof. Suppose $\mathrm{pK}^t_\alpha(x) = k$. Then by definition, we have

$$\Pr_{w\sim\{0,1\}^t}[\exists M_w \in \{0,1\}^k \text{ such that } M_w(w) \text{ outputs } x \text{ within } t \text{ steps}] \geq \alpha.$$

Call such a w good. After sampling ℓ independent $w^{(1)}, w^{(2)}, \ldots, w^{(\ell)} \in \{0,1\}^t$, the probability that no $w^{(i)}$ is good is at most $(1-\alpha)^\ell \leq e^{-\alpha\ell}$, which can be made at most $1-\beta$ by choosing $\ell = q/\alpha$. It follows that, with probability at least β over a random $w = (w^{(1)}, w^{(2)}, \ldots, w^{(q/\alpha)}) \in \{0,1\}^{qt/\alpha}$, there exists an index $1 \leq i \leq q/\alpha$ (described with at most $\lceil \log(q/\alpha) \rceil$ bits) and a program $M_{w^{(i)}} \in \{0,1\}^k$ such that $M_{w^{(i)}}(w^{(i)})$ outputs x within $O(qt/\alpha)$ steps. Hence, $\mathrm{pK}^{O(qt/\alpha)}_\beta(x) \leq k + O(\log(q/\alpha))$. ◁ Note that the parameter α affects the time complexity in the resulting $\mathrm{pK}^{O(qt/\alpha)}_\beta$ bound. Lemma 21 is sufficient in applications where $\alpha \geq 1/\operatorname{poly}(n)$.

3.2 Bounding pK^t via DPG reconstruction (selected original case)

The following is a key lemma for pK^t that we will use in proving our main results related to randomized average-case complexity.

Lemma 22 (pK^t Reconstruction Lemma). For $\varepsilon > 0$, $x \in \{0,1\}^n$, $s \in \mathbb{N}$, and $k \in \mathbb{N}$ satisfying $k \leq 2n$, let D be a randomized algorithm that takes an advice string β and runs in time t_D such that D ε-distinguishes $\mathrm{DP}_k(x; U_{nk})$ from U_{nk+k}. Then there is a polynomial p_{DP} such that

$$\mathrm{pK}^{\widetilde{O}(t_D)\cdot p_{\mathrm{DP}}(n/\varepsilon)}(x \mid \beta) \leq k + \log p_{\mathrm{DP}}(nt_D/\varepsilon).$$

[4] We use the RAM model in this definition for convenience: it simplifies the time bound estimates for a machine M that takes as input both randomness w and an "auxiliary" string y, and allows both w and y to be as long as the running time of the RAM-machine M; cf. [29] for a similar definition in the case of deterministic time-bounded conditional Kolmogorov complexity. Alternatively, we could use a TM model, with inputs w and y presented on two different tapes. However, we prefer the RAM model as it's also useful in the context of fine-grained worst-case to average-case reductions [6].

Proof. We will view the distinguisher D as a deterministic algorithm that takes a string of t_D bits, denoted by r_D, as its internal randomness. By our assumption (and dropping the absolute value sign without loss of generality), we have

$$\mathbb{E}_{y\sim\{0,1\}^{nk},r_D\sim\{0,1\}^{t_D}}[D(\mathrm{DP}_k(x,y);r_D)]-\mathbb{E}_{z\sim\{0,1\}^{nk+k},r_D\sim\{0,1\}^{t_D}}[D(z;r_D)]\geq\varepsilon.$$

Then by an averaging argument, we have

$$\Pr_{r_D}[\mathbb{E}_y[D(\mathrm{DP}_k(x,y);r_D)]-\mathbb{E}_z[D(z;r_D)]\geq\varepsilon/2]\geq\varepsilon/2. \tag{4}$$

In other words, with probability at least $\varepsilon/2$ over the internal randomness r_D, $D(-,r_D)$ is an $(\varepsilon/2)$-distinguisher for $\mathrm{DP}_k(x;U_{nk})$ and U_{nk+k}. Let us say that r_D is good if this is true. Let $\mathrm{Recon}^{(-)}:\{0,1\}^k\times\{0,1\}^r\to\{0,1\}^n$ be the reconstruction procedure in Lemma 14 that works for distinguishing parameter $\varepsilon/2$, where the randomness complexity r is $\mathrm{poly}(n/\varepsilon)$. We have

$$\begin{aligned}
&\Pr_{w\sim\{0,1\}^r,r_D\sim\{0,1\}^{t_D}}[\exists\alpha\in\{0,1\}^k \text{ such that } \mathrm{Recon}^{D(-,r_D)}(\alpha,w)=x]\\
&\geq \Pr_{w,r_D}[\exists\alpha\in\{0,1\}^k \text{ such that } \mathrm{Recon}^{D(-,r_D)}(\alpha,w)=x \mid r_D \text{ is good}]\cdot\Pr_{r_D}[r_D \text{ is good}]\\
&\geq \frac{1}{\mathrm{poly}(n/\varepsilon)}\cdot(\varepsilon/2). \qquad \text{(Lemma 14 and Equation (4))}
\end{aligned}$$

The above implies that for at least $1/\mathrm{poly}(n/\varepsilon)$ fraction of the randomness (w,r_D), there exists a program M, which takes k bits for some α (that can depend on the randomness) and the advice of β used by D, such that M simulates $\mathrm{Recon}^{D(-,r_D)}$ on (α,w) and outputs x. Note that since Recon^D runs in time $t_D\cdot\mathrm{poly}(n/\varepsilon)$, our program M can be made to run in time $T:=\widetilde{O}(t_D)\cdot\mathrm{poly}(n/\varepsilon)$. Therefore, we have

$$\mathrm{pK}^T_{1/\mathrm{poly}(n/\varepsilon)}(x\mid\beta)\leq k+O(\log(nt_D/\varepsilon)).$$

Then the lemma follows easily from Lemma 21. ◁

3.4 Symmetry of information (original Item1 and its full proof)

Lemma 26 (Symmetry of information for pK^t and rK^t under the average-case easiness of NP), Item 1. If DistNP $\subseteq$ AvgBPP, then there exist polynomials p and p_0 such that for all sufficiently large $n,m\in\mathbb{N}$, all $t\geq p_0(n,m)$, and all $0\leq\tau\leq t$ the following holds: for every $x\in\{0,1\}^n$ and every family $\{y_r\in\{0,1\}^m\mid r\in\{0,1\}^\tau\}$, with probability at least $9/10$ over $r\in\{0,1\}^\tau$,

$$\mathrm{pK}^t(x,y_r\mid r)>\mathrm{pK}^{p(t)}(x\mid r)+\mathrm{pK}^{p(t)}(y_r\mid x,r)-\log p(t).$$

Moreover, the special case without conditioning on r also holds for both items above, i.e., when r is the empty string (for $\tau=0$) and y is an arbitrary single string.

Proof of Item 1. Define a language

$$L := \{(u, v, w, w', 1^s) \mid \exists M \in \{0,1\}^s, M(w, w') \text{ prints } uv \text{ in } |w| \text{ steps, and } s = |u|+|v|-10\}.$$

Note that $L \in$ NP. Define a distribution family $\mathcal{D} = \{D_{\langle nk+k, mk'+k', 2t, \tau, s\rangle}\}$ as follows: sample $u \sim U_{nk+k}$, $v \sim U_{mk'+k'}$, $w \sim U_{2t}$, and $w' \sim U_\tau$, and then output $(u, v, w, w', 1^s)$. Note that the ensemble $\mathcal{D}$ is in PSamp under an appropriate encoding of its defining tuple (for simplicity, we omit the padding in the output of $\mathcal{D}$; see Section 2.1). Using the assumption that DistNP $\subseteq$ AvgBPP, it follows that $(L, \mathcal{D}) \in$ AvgBPP. Let B be a randomized algorithm for $(L, \mathcal{D})$ as in Lemma 6. Let $x \in \{0,1\}^n$ and $\{y_r\}_r \subseteq \{0,1\}^m$ be given (for sufficiently large n and m). Let $k, k' \in \mathbb{N}$ be arbitrary parameters such that $k \leq n$ and $k' \leq m$. Observe that there exists a polynomial p_0 such that for any $t \geq p_0(n, m)$, some constant d, and all $z \in \{0,1\}^{nk}$, $z' \in \{0,1\}^{mk'}$, and $r \in \{0,1\}^\tau$,

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) \leq \mathrm{pK}^t(x, y_r \mid r) + |z| + |z'| + d \log t. \tag{6}$$

Here $p_0(n, m)$ reflects the time required to deterministically compute $(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'))$ given x, y_r, z, z', and $d \log t$ bits of information to delineate x from y_r. Let $t \geq p_0(n, m)$. By a counting argument, for independently random u, v, w, and w',

$$\Pr_{u,v,w,w'}[(u, v, w, w', 1^s) \in L] \leq \frac{2^s \cdot 2^{|w|+|w'|}}{2^{|u|+|v|+|w|+|w'|}} = 2^{s-|u|-|v|} = 2^{-10},$$

where the last equality is by the definition of L requiring that $s = |u| + |v| - 10$. Then

$$\begin{aligned}\Pr_{u,v,w,w',r_B}[B(u, v, w, w', 1^s) = 1] \leq &\Pr_{u,v,w,w'}[(u, v, w, w', 1^s) \in L] \\ &+ \Pr_{u,v,w,w',r_B}[B(u, v, w, w', 1^s) \neq L(u, v, w, w', 1^s)] \\ &\leq 2^{-10} + (3/n). \end{aligned} \tag{7}$$

By Markov's inequality, for at least 9/10 fraction of random strings $r \in \{0,1\}^\tau$, we get

$$\Pr_{u,v,w,r_B}[B(u, v, w, r, 1^s) = 1] \leq 10 \cdot (2^{-10} + (3/n)) \leq 1/10. \tag{8}$$

Fix any such r so that Equation (8) holds. This also fixes the string y_r. Next we show, using a hybrid argument, that $B(-, U_{2t}, r, 1^s)$ cannot distinguish between the uniform distribution and the distribution $(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'))$, for random independent z, z', where $k \approx \mathrm{pK}^{p_{\mathrm{DP}}(t)}(x \mid r)$ and $k' \approx \mathrm{pK}^{p_{\mathrm{DP}}(t)}(y_r \mid x, r)$. This will imply that

$$\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] < 2/3$$

for some z, z', yielding the desired lower bound on $\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r)$ via our choice of s and the definition of L. We give the details of the hybrid argument next. First, toward a contradiction, suppose

$$\Pr_{z,v,w,r_B}[B(\mathrm{DP}_k(x; z), v, w, r, 1^s) = 1] > 1/2.$$

In this case, comparing with Equation (8), we get a randomized distinguisher (with advice) for $\mathrm{DP}_k(x; U_{nk})$, defined by sampling $v \sim U_{mk'+k'}$ and $w \sim U_{2t}$, and outputting $B(-, v, w, r, 1^s)$. By Lemma 22,

$$\mathrm{pK}^{q(t)}(x \mid r) \leq k + \log q(t) \tag{9}$$

for some polynomial q such that $q(t) \geq p_{\mathrm{DP}}(t_B \cdot 3 \cdot n)$ whenever $t \geq n + m$ and $s \leq n^3$, where p_{DP} is the polynomial from Lemma 22 and t_B denotes the time required to compute $B(-, v, w, r, 1^s)$. Define $k := \mathrm{pK}^{q(t)}(x \mid r) - \log q(t) - 1$ so that Equation (9) does not hold. Assume for now that $k > 0$. Hence,

$$\Pr_{z,v,w,r_B}[B(\mathrm{DP}_k(x; z), v, w, r, 1^s) = 1] \leq 1/2. \tag{10}$$

Again, toward a contradiction, suppose that for all z, z',

$$\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] \geq 2/3.$$

By definition of B, this implies that

$$\Pr_{z,z',w,r_B}[B(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) = 1] \geq (2/3)(1 - 1/n) > 5/8.$$

In this case, comparing with Equation (10), we get a randomized distinguisher (with advice) B' for $\mathrm{DP}_{k'}(y_r; U_{mk'})$, defined by sampling $z \sim U_{nk}$, $w \sim U_{2t}$, and outputting $B(\mathrm{DP}_k(x; z), -, w, r, 1^s)$. By Lemma 22,

$$\mathrm{pK}^{q'(t)}(y_r \mid x, r) \leq k' + \log q'(t) \tag{11}$$

for some polynomial q' with $q'(t) \geq p_{\mathrm{DP}}(t_{B'} \cdot 8 \cdot m)$ whenever $t \geq n + m$ and $s \leq n^3$, where $t_{B'}$ denotes the time required to compute B'. We now choose $k' := \mathrm{pK}^{q'(t)}(y_r \mid x, r) - \log q'(t) - 1$ so that Equation (11) does not hold. Assume for now that $k' > 0$. Hence, there exist z and z' such that

$$\begin{aligned}&\Pr_w[(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z'), w, r, 1^s) \in L] \\ &= \Pr_w[\exists M \in \{0,1\}^s, M(w, r) \text{ outputs } (\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z')) \text{ within } |w| \text{ steps}] < 2/3,\end{aligned}$$

which implies that

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) > s.$$

For this choice of z, z', by definition of s we get that $s = |u| + |v| - 10 = |z| + k + |z'| + k' - 10$, and so

$$\mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) > |z| + k + |z'| + k' - 10.$$

Combining this inequality with Equation (6), we get

$$\begin{aligned}\mathrm{pK}^t(x, y_r \mid r) &\geq \mathrm{pK}^{2t}(\mathrm{DP}_k(x; z), \mathrm{DP}_{k'}(y_r; z') \mid r) - |z| - |z'| - d \log t \\ &> k + k' - d \log t - 10.\end{aligned} \tag{12}$$

The definitions of k and k' then imply that

$$\begin{aligned}\mathrm{pK}^t(x, y_r \mid r) > \mathrm{pK}^{q(t)}(x \mid r) + \mathrm{pK}^{q'(t)}(y_r \mid x, r) \\ - \log q(t) - \log q'(t) - d \log t - 12.\end{aligned}$$

For the polynomial $p(t) := (t^{d+1}) \cdot q(t) \cdot q'(t)$ and for every $t \geq p_0(|x|, |y|)$, we get

$$\mathrm{pK}^t(x, y_r \mid r) > \mathrm{pK}^{p(t)}(x \mid r) + \mathrm{pK}^{p(t)}(y_r \mid x, r) - \log p(t),$$

as desired. Finally, consider the case that $k \leq 0$ or $k' \leq 0$. If $k \leq 0$, then $\mathrm{pK}^{q(t)}(x \mid r) \leq \log q(t) + 1$, implying that $\mathrm{pK}^{p(t)}(x \mid r) < \log p(t)$. But then the lemma simply follows from the fact that $\mathrm{pK}^t(x, y_r \mid r) \geq \mathrm{pK}^{p(t)}(y_r \mid x, r)$. Similarly, if $k' \leq 0$, then $\mathrm{pK}^{p(t)}(y_r \mid x, r) < \log p(t)$, and the lemma follows from the fact that $\mathrm{pK}^t(x, y_r \mid r) \geq \mathrm{pK}^{p(t)}(x \mid r)$. ◁

3.7 Probabilistic computational depth (original Item1 and its full proof)

The next lemma is a straightforward adaptation of an argument from [11]. We provide a proof for completeness.

Lemma 31 (Computational Depth Upper Bound), Item 1. For all $x \in \{0,1\}^n$ and $r \in \{0,1\}^*$, and for all non-decreasing polynomials p_0 and p, if n is large enough then there exists a time bound t_1 such that $p_0(n) \leq t_1 \leq 2^{n/\log n}$ and

$$\mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{p(t_1)}(x \mid r) \leq O(n/\log n).$$

Proof. We will start by proving Item 1. Given $x \in \{0,1\}^n$ and polynomials p_0 and p, define the polynomial $\tau := p \circ p_0$. For an integer $I \geq 1$, consider the following telescoping sum:

$$\begin{aligned}\mathrm{pK}^{\tau(n)}(x \mid r) - \mathrm{pK}^{\tau^{I+1}(n)}(x \mid r) = {} & (\mathrm{pK}^{\tau(n)}(x \mid r) - \mathrm{pK}^{\tau^2(n)}(x \mid r)) \\ & + (\mathrm{pK}^{\tau^2(n)}(x \mid r) - \mathrm{pK}^{\tau^3(n)}(x \mid r)) \\ & + \cdots + (\mathrm{pK}^{\tau^I(n)}(x \mid r) - \mathrm{pK}^{\tau^{I+1}(n)}(x \mid r)),\end{aligned}$$

where $\tau^i(-)$ denotes the composition of τ with itself i times. For any choice of x, p_0, and p as in the statement of the lemma, $\mathrm{pK}^{\tau(n)}(x \mid r) \leq n + d$, for some universal constant $d \geq 0$; hence, the above sum is at most $n + d$. By averaging, there is some index $i_0 \in [I]$ such that

$$\mathrm{pK}^{\tau^{i_0}(n)}(x \mid r) - \mathrm{pK}^{\tau^{i_0+1}(n)}(x \mid r) \leq (n+d)/I.$$

For this i_0, define $t_1 := \tau^{i_0}(n)$. Note that $t_1 \geq \tau(n) \geq p_0(n)$, since $i_0 \geq 1$ and $p(\ell) \geq \ell$ for every input ℓ. Letting $c \in \mathbb{N}$ be such that $\tau(n) \leq n^c$ for sufficiently large n, define $I := \log_c(n/(\log_2 n)^2)$. Then $t_1 \leq n^{c^I} = 2^{n/\log n}$. Moreover,

$$\begin{aligned}\mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{p(t_1)}(x \mid r) &\leq \mathrm{pK}^{t_1}(x \mid r) - \mathrm{pK}^{\tau(t_1)}(x \mid r) \\ &\leq O(n/\log n),\end{aligned}$$

as desired.

2.6 Agnostic learning model (author definitions)

Consider the problem of learning an unknown concept $f : X \to \{0,1\}$ over a finite domain X. We will always let $X = \{0,1\}^n$, for some $n \geq 1$. A concept class $\mathcal{C}$ is a collection of concepts of this form. For a fixed n, we let $\mathcal{C}_n$ denote $\mathcal{C} \cap \{f : \{0,1\}^n \to \{0,1\}\}$. For a size function $s : \mathbb{N} \to \mathbb{N}$, we let

$$\text{SIZE}[s] = \{f : \{0,1\}^n \to \{0,1\} \mid n \in \mathbb{N} \text{ and } f \text{ admits a size-}s(n) \text{ circuit}\}$$

denote the concept class of Boolean circuits of size (number of gates) at most $s(n)$ over n input variables. For definiteness, we consider circuits over AND and OR gates of fan-in 2 and NOT gates. A randomized Boolean function f maps an input x to a distribution D supported over $\{0,1\}$. Given a distribution D_n supported over $\{0,1\}^n$, a function $h : \{0,1\}^n \to \{0,1\}$, and a (possibly randomized) Boolean function $f : \{0,1\}^n \to \{0,1\}$, we let

$$\text{err}_{D_n}(h,f) = \Pr_{f,x\sim D_n}[h(x) \neq f(x)].$$

We will consider the (agnostic) learnability of a concept class $\mathcal{C}$ with respect to a class $\mathcal{D}$ of ensembles $D = \{D_n\}_{n\geq 1}$ of distributions D_n. We use $\mathcal{D}_n$ to denote $\{D_n \mid D \in \mathcal{D}\}$. For simplicity, we refer to $\mathcal{D}$ as a class of distributions. Given a randomized function $f : \{0,1\}^n \to \{0,1\}$ and a distribution D_n, the learning algorithm has access to an example oracle $\text{EX}(f, D_n)$ that behaves as follows: each query to $\text{EX}(f, D_n)$ returns an independent and identically distributed pair (x, b), where x is sampled according to D_n and $b = f(x)$. The sample complexity of the learning algorithm is the number of queries made to $\text{EX}(f, D_n)$. Before formalising the agnostic learning model, we provide an informal overview. The learning algorithm knows the class $\mathcal{D}$ of distributions and the concept class $\mathcal{C}$. It is given access to $\text{EX}(f, D_n)$ for some unknown $D \in \mathcal{D}_n$ and an arbitrary (possibly randomized) function $f : \{0,1\}^n \to \{0,1\}$. The goal of the learning algorithm is to produce, with high probability over its internal randomness and queries to $\text{EX}(f, D)$, a hypothesis h such that $\text{err}_D(h,f) \leq \text{opt}_{\mathcal{C}_n,D,f} + \varepsilon$, where

$$\text{opt}_{\mathcal{C}_n,D,f} = \min_{c\in\mathcal{C}_n} \text{err}_D(c,f).$$

In other words, the learning algorithm is required to output an efficient representation of a hypothesis $h : \{0,1\}^n \to \{0,1\}$ that is almost as accurate as the best concept in $\mathcal{C}_n$ with respect to the pair (D, f). (Note that h is not required to be a function in $\mathcal{C}$.)

Definition 15 (Agnostic PAC learning [20]). Let $\mathcal{C}$ be a concept class, and let $\mathcal{D}$ be a class of distributions. We say that a randomized algorithm A agnostically learns $\mathcal{C}$ on $\mathcal{D}$ if the following holds. For every $n \geq 1$, distribution $D \in \mathcal{D}_n$, and randomized function $f : \{0,1\}^n \to \{0,1\}$, when A is given oracle access to $\text{EX}(f, D)$ and receives as input $n, \varepsilon > 0$, and $\delta > 0$,

$$\Pr_{A,\text{EX}(f,D)}[A^{\text{EX}(f,D)} \text{ outputs a hypothesis } h \text{ such that } \text{err}_D(h,f) \leq \text{opt}_{\mathcal{C}_n,D,f} + \varepsilon] \geq 1-\delta.$$

We remark that an equivalent way of formulating the agnostic learning model is by considering distributions D supported over $\{0,1\}^{n+1}$. In this case, there is no need to refer to randomized functions, and one measures the error of a hypothesis h via $\text{err}_D(h) = \Pr_{(x,b)\sim D}[h(x) \neq b]$, where $x \in \{0,1\}^n$ and $b \in \{0,1\}$.

Additional author pKt properties

Lemma 18. There is a universal constant $c > 0$ such that the following holds. For every time bound $t \in \mathbb{N}$ and $x \in \{0,1\}^n$,

$$\mathrm{K}(x \mid t) \leq \mathrm{pK}^t(x) + c \log n.$$

Proof. Let $\ell := \mathrm{pK}^t(x)$. Since x is of length n, we can represent ℓ using $O(\log n)$ bits. Given t, ℓ and n, it is possible to sample x with probability $\mu \geq \frac{2}{3} \cdot 2^{-\ell}$ by randomly guessing $w \sim \{0,1\}^t$ and $M \sim \{0,1\}^\ell$ then simulating $M(w)$ for t steps. Consequently, by the coding theorem for (time-unbounded) Kolmogorov complexity [24], we get that $\mathrm{K}(x \mid t) \leq \log(1/\mu) + O(\log n) \leq \mathrm{pK}^t(x) + O(\log n)$. ◁

Lemma 20 (Probabilistic Incompressibility). For any string $y \in \{0,1\}^*$, time bound $t \in \mathbb{N}$ (including $t = \infty$), $\delta \in (0,1]$, and positive integer α, we have

$$\Pr_{x \sim \{0,1\}^n}(\mathrm{pK}^t_\delta(x \mid y) \leq n - \alpha) \leq \frac{1}{2^{\alpha-1} \cdot \delta}.$$

Proof. For the sake of contradiction, suppose the statement of the lemma does not hold. Then by the definition of pK^t_δ, we have

$$\mathbb{E}_{x \sim \{0,1\}^n, w \sim \{0,1\}^t}[\exists M_{(x,w)} \in \{0,1\}^{\leq n-\alpha} \text{ such that } M_{(x,w)}(w,y) \text{ outputs } x \text{ within } t \text{ steps}] > 1/2^{\alpha-1}.$$

By averaging, there is a way to fix $w \in \{0,1\}^t$ that at least preserves this expectation, yielding

$$\Pr_{x \sim \{0,1\}^n}[\exists M_{(x,w)} \in \{0,1\}^{\leq n-\alpha} \text{ such that } M_{(x,w)}(w,y) \text{ outputs } x \text{ within } t \text{ steps}] > 1/2^{\alpha-1}.$$

However, by counting, the above probability is at most $2^{n-\alpha+1}/2^n = 1/2^{\alpha-1}$. A contradiction. ◁

Lemma 24 (Unconditional Source Coding for pK^t [34]). There exists a polynomial p such that for any $T, a : \mathbb{N} \to \mathbb{N}$, $n \in \mathbb{N}$, $D_n \in \mathrm{Samp}[T(n)]/a(n)$, and $x \in \mathrm{Supp}(D_n)$,

$$\mathrm{pK}^{p(T(n))}(x) \leq \log(1/D_n(x)) + O(\log(T(n))) + a(n).$$

For a distribution D_n supported over X and an integer $m \geq 1$, we let $D_n^m := D_n \times \ldots \times D_n$ be the probability distribution supported over X^m obtained by taking the product of m independent copies of D_n. As a consequence of Lemma 24, we obtain the following result.

Lemma 25 (Source Coding for m copies of D_n). There exists a polynomial p such that for any $T, a : \mathbb{N} \to \mathbb{N}$, $n, m \in \mathbb{N}$, $D_n \in \mathrm{Samp}[T(n)]/a(n)$, and $x \in \mathrm{Supp}(D_n^m)$,

$$\mathrm{pK}^{p(T(n),m)}(x) \leq \log(1/D_n^m(x)) + O(\log(T(n))) + a(n) + O(\log(m)).$$

Proof. Since D_n can be sampled in time $T(n)$ using $a(n)$ bits of advice, it is possible to sample from D_n^m in time $\mathrm{poly}(T(n), m)$ using $a(n) + O(\log m)$ bits of advice, where the extra advice is used to encode m. The result now immediately follows from Lemma 24. ◁

3.5 Approximating pK^t under randomized average-case easiness

Lemma 27. If DistNP $\subseteq$ AvgBPP, then there exists a polynomial τ such that the following promise problem is in promiseBPP:

$$\Pi_{\mathrm{YES}} := \{(x, 1^s, 1^t) \mid \mathrm{pK}^t(x) \leq s\},$$
$$\Pi_{\mathrm{NO}} := \{(x, 1^s, 1^t) \mid \mathrm{pK}^{\tau(t)}(x) > s + \log \tau(t)\}.$$

Proof. Let $k = s + \log n$. Consider the language

$$L := \{(\mathrm{DP}_k(x; z), w, 1^s, 1^n) \mid \exists M \in \{0,1\}^s, M(w) \text{ outputs } x \in \{0,1\}^n \text{ within } |w| \text{ steps}\}.$$

Note that $L \in$ NP. Define a distribution family $\mathcal{D} = \{D_{\langle nk+k,t,s,n\rangle}\}$, each member of which does the following: sample $u \sim U_{nk+k}$ and $w \sim U_t$, and then output $(u, w, 1^s, 1^n)$. By assumption, $(L, \mathcal{D}) \in$ AvgBPP. Let B be a randomized heuristic algorithm for $(L, \mathcal{D})$ as described in Lemma 6. Now, define an algorithm B':

- On input $(x, 1^s, 1^t)$ with $x \in \{0,1\}^n$, sample $z \sim U_{nk}$ and $w \sim U_t$, and then output $B(\mathrm{DP}_k(x; z), w, 1^s, 1^n)$.

Below, we show that B' solves $(\Pi_{\mathrm{YES}}, \Pi_{\mathrm{NO}})$ correctly with high probability in the worst case. First, consider the case that $(x, 1^s, 1^t) \in \Pi_{\mathrm{YES}}$. By the definitions of L and pK, for any choice of $z \in \{0,1\}^{nk}$,

$$\Pr_w[(\mathrm{DP}_k(x; z), w, 1^s, 1^n) \in L] \geq 2/3.$$

The definition of B then implies that

$$\Pr_{w,z,r_B}[B(\mathrm{DP}_k(x; z), w, 1^s, 1^n) = 1] > 1/2,$$

and so

$$\Pr_{r_{B'}}(B'(x, 1^s, 1^t) = 1) > 1/2, \tag{13}$$

where r_B denotes the internal randomness of B, and $r_{B'} = (w, z, r_B)$ that of B'. Now consider the case that $(x, 1^s, 1^t) \in \Pi_{\mathrm{NO}}$. For a contradiction, suppose that

$$\Pr_{w,z,r_B}[B(\mathrm{DP}_k(x; z), w, 1^s, 1^n) = 1] > 1/3. \tag{14}$$

By a counting argument, for randomly selected u and w,

$$\Pr_{u,w}[(u, w, 1^s, 1^n) \in L] \leq \frac{2^s \cdot 2^{nk} \cdot 2^{|w|}}{2^{nk+k+|w|}} = 1/n,$$

where the last line follows from the definition of $k = s + \log n$. Then by definition of B,

$$\Pr_{u,w,r_B}[B(u, w, 1^s, 1^n) = 1] \leq 4/n = o(1). \tag{15}$$

Comparing Equations (14) and (15), it is clear that $B(-, U_t, 1^s, 1^n)$ (1/4)-distinguishes $\mathrm{DP}_k(x; U_{nk})$ from U_{nk+k}. Lemma 22 implies that

$$\mathrm{pK}^{p'(t)}(x) \leq k + \log p'(t) = s + \log n + \log p'(t),$$

for some polynomial p' with $p'(t) \geq p_{\mathrm{DP}}(t_B \cdot 4 \cdot n)$, where p_{DP} is the polynomial from Lemma 22 and t_B denotes the time required to compute $B(-, U_t, 1^s, 1^n)$. For the polynomial $\tau(t) = t \cdot p'(t)$, this means that $(x, 1^s, 1^t)$ is not in Π_{NO}, which gives the desired contradiction. By definition of B', we have that

$$\Pr_{r_{B'}}(B'(x, 1^s, 1^t) = 1) \leq 1/3. \tag{16}$$

By Equations (13) and (16), B' yields a promiseBPP algorithm for $(\Pi_{\mathrm{YES}}, \Pi_{\mathrm{NO}})$ via standard success amplification techniques. ◁

Lemma 28 (Approximating pK^t). If DistNP $\subseteq$ AvgBPP, then there exist a polynomial τ, a constant $C \geq 1$, and a randomized algorithm Approx$_\tau$-pK that, on input $(x, 1^t)$ where $x \in \{0,1\}^n$ and $t \geq Cn$, runs in time $\mathrm{poly}(n, t)$ and with probability at least $1 - o(1)$ outputs an integer $\widetilde{s}$ such that

$$\mathrm{pK}^{\tau(t)}(x) - \log \tau(t) \leq \widetilde{s} \leq \mathrm{pK}^t(x).$$

Proof. Consider a polynomial-time randomized algorithm A that solves the promise problem from Lemma 27. Assume without loss of generality that on the inputs satisfying the promise its error is at most $1/n^2$, where $n = |x|$. Algorithm Approx$_\tau$-pK runs A on $(x, 1^s, 1^t)$ for $s = 1, 2, \ldots, n + \log n$, and outputs the first $\widetilde{s}$ such that $A(x, 1^{\widetilde{s}}, 1^t) = 1$. The correctness of Approx$_\tau$-pK follows by a union bound. Indeed, if $s < \mathrm{pK}^{\tau(t)}(x) - \log \tau(t)$, i.e., $\mathrm{pK}^{\tau(t)}(x) > s + \log \tau(t)$, using the promise we get that $\Pr_A[A(x, 1^s, 1^t) = 1] \leq 1/n^2$. On the other hand, if $s = \mathrm{pK}^t(x)$, which implies that $\mathrm{pK}^t(x) \leq s$ and the promise is satisfied, we have $\Pr_A[A(x, 1^s, 1^t) = 1)] \geq 1 - 1/n^2$. Since $\mathrm{pK}^t(x) \leq n + \log n$ if $t \geq Cn$, where C is a sufficiently large constant, with high probability over the internal randomness of Approx$_\tau$-pK, it outputs a value $\widetilde{s}$ such that $\mathrm{pK}^{\tau(t)}(x) - \log \tau(t) \leq \widetilde{s} \leq \mathrm{pK}^t(x)$. ◁

5 Learning in Randomized Heuristica

The main result of this section is the following.

Theorem 56. If DistNP $\subseteq$ AvgBPP, then for any time constructible functions $s, T, a : \mathbb{N} \to \mathbb{N}$, and $\varepsilon \in [0, 1]$, SIZE$[s(n)]$ is agnostic learnable on Samp$[T(n)]/a(n)$ in time $\mathrm{poly}(n, \varepsilon^{-1}, s(n), T(n), a(n))$ with sample complexity

$$\left(\frac{(n + s(n) + a(n) + \log T(n))^3}{\varepsilon^8}\right)^{1+o(1)}.$$

5.1 Ingredients

In this section, we will use D to denote a family of distributions (one for each n) and $\mathcal{D}$ to denote a class of family of distributions (e.g., those that are efficiently samplable). For a distribution D and $m \in \mathbb{N}$, let D^m denote the distribution $x_1 \circ x_2 \circ \cdots \circ x_m$ where $x_1, x_2, \ldots, x_m \sim D$.

5.1.1 Learning from RRHS Refutation

Definition 57 (Correlative RRHS-Refutation [21, Definition 3]). Let $\mathcal{C}$ be a concept class, and let $\mathcal{D}$ be a class of example distributions. A randomized algorithm A is a correlatively random-right-hand-side-refuter (correlative RRHS-refuter) for $\mathcal{C}$ on $\mathcal{D}$ with sample complexity m if A satisfies the following. A takes as input $n \in \mathbb{N}$, $\varepsilon \in (0,1)$ and a set $S = (\langle x^{(1)}, b^{(1)} \rangle, \ldots, \langle x^{(m)}, b^{(m)} \rangle)$ of samples, where $x^{(i)} \in \{0,1\}^n$ and $b^{(i)} \in \{0,1\}$ for every $i \in [m]$, and

- Soundness: if the samples S are i.i.d. from a distribution D' on $\{0,1\}^n \times \{0,1\}$ such that the marginal on $\{0,1\}^n$ equals D_n for some $D_n \in \mathcal{D}_n$ and there exists $f \in \mathcal{C}_n$ with

$$\Pr_{\langle x^{(i)}, b^{(i)} \rangle \sim D'}[b^{(i)} = f(x^{(i)})] \geq \frac{1}{2} + \frac{\varepsilon}{2},$$

  then $\Pr_{S,A}[A(n, \varepsilon, S) = \text{correlative}] \geq 2/3$.

- Completeness: if the samples S are selected i.i.d. so that $x^{(1)}, \ldots, x^{(m)} \sim D_n$ for some $D_n \in \mathcal{D}_n$ and $b^{(1)}, \ldots, b^{(m)} \sim U$, then

$$\Pr_{S,A}[A(n, \varepsilon, S) = \text{random}] \geq 2/3.$$

Theorem 58 (Agnostic Learning from RRHS-Refutation [21]; see also [13, Theorem 5]). Let $\mathcal{C}$ be a concept class, and let $\mathcal{D}$ be a class of example distributions. If there exists a correlative RRHS-refuter for $\mathcal{C}$ on $\mathcal{D}$ with sample complexity $m(n, \varepsilon)$ and running time $T(n, \varepsilon)$, then $\mathcal{C}$ is agnostic learnable with

$$\text{sample complexity} = O\left(\frac{m(n, \varepsilon/2)^3}{\varepsilon^2}\right) \text{ and}$$
$$\text{running time} = O\left(T(n, \varepsilon/2) \cdot \frac{m(n, \varepsilon/2)^2}{\varepsilon^2}\right).$$

5.1.2 Probabilistic sampling depth

Definition 59 (Probabilistic Sampling Depth). Let $t, t' \in \mathbb{N}$, where $t' > t$. For a family $D = \{D_n\}_{n \geq 1}$ of distributions, we introduce the (t, t')-probabilistic-sampling-depth functions $\mathrm{psd}_D^{t,t'} := \{\mathrm{psd}_{D,n}^{t,t'}\}_{n \geq 1}$, where

$$\mathrm{psd}_{D,n}^{t,t'}(m) = \mathbb{E}_{X \sim D_n^m}[\mathrm{pK}^t(X) - \mathrm{pK}^{t'}(X)].$$

For a collection $\mathcal{D}$ of families of distributions, we define $\mathrm{psd}_{\mathcal{D}}^{t,t'} := \{\mathrm{psd}_{\mathcal{D},n}^{t,t'}\}_{n \geq 1}$, where

$$\mathrm{psd}_{\mathcal{D},n}^{t,t'}(m) = \max_{D \in \mathcal{D}} \mathrm{psd}_{D,n}^{t,t'}(m).$$

Lemma 60 (Small Probabilistic Sampling Depth for Samplable Distributions). There exists a polynomial $p_1 : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that for any $T, a : \mathbb{N} \to \mathbb{N}$ and $n, m \in \mathbb{N}$, the following holds. For every $t \geq p_1(T(n), m)$, and every $t' > t$,

$$\mathrm{psd}^{t,t'}_{\mathrm{Samp}[T(n)]/a(n),n}(m) \leq O(\log m + \log T(n) + a(n) + \log t').$$

Proof. The proof is essentially the same as that of [13, Lemma 6], but uses the unconditional coding theorem for probabilistic Kolmogorov complexity (Lemma 25). Fix any $D_n \in \mathrm{Samp}[T(n)]/a(n)$. Let p_1 be the polynomial p in Lemma 25. We have for every $t \geq p_1(T(n), m)$,

$$\begin{aligned}
\mathbb{E}_{x\sim D_n^m}[\mathrm{pK}^t(x)] &\leq \mathbb{E}_{x\sim D_n^m}[\mathrm{pK}^{p_1(T(n),m)}(x)] \\
&\leq \mathbb{E}_{x\sim D_n^m}[\log(1/D_n^m(x))] + O(\log(m) + \log T(n) + a(n)) \quad \text{(Lemma25)} \\
&= H(D_n^m) + O(\log(m) + \log T(n) + a(n)) \\
&\leq \mathbb{E}_{x\sim D_n^m}[\mathrm{K}(x)] + O(\log(m) + \log T(n)) + a(n) \\
&\leq \mathbb{E}_{x\sim D_n^m}[\mathrm{pK}^{t'}(x)] + O(\log(m) + \log T(n) + a(n) + \log t') \quad \text{(Lemma18)},
\end{aligned}$$

where the second last inequality uses the fact that for any distribution D, its Shannon entropy $H(D) \leq \mathbb{E}_{y\sim D}[\mathrm{K}(y)]$ (see [27, Theorem 8.1.1] and [13, Lemma 5]). Rearranging the above, we get

$$\mathbb{E}_{x\sim D_n^m}[\mathrm{pK}^t(x) - \mathrm{pK}^{t'}(x)] \leq O(\log(m) + \log T(n) + a(n) + \log t'),$$

as desired. ◁

5.2 RRHS refuters from probabilistic average-case easiness

We first prove the following technical theorem.

Theorem 61. If DistNP $\subseteq$ AvgBPP, then for any time constructible functions $s, T, a : \mathbb{N} \to \mathbb{N}$ and for any constant $\zeta > 0$, there is a correlative RRHS-refuter for SIZE$[s(n)]$ under Samp$[T(n)]/a(n)$ with sample complexity

$$m := \left(\frac{n + s(n) + a(n) + \log T(n)}{\varepsilon^2}\right)^{1+\zeta}$$

and running time $\mathrm{poly}(n, m, T(n)) = \mathrm{poly}(n, 1/\varepsilon, T(n), a(n), s(n))$.

Proof. The proof closely follows that of [13, Theorem 8].

The (correlative) RRHS-refuter R. Let τ be the polynomial from Lemma 28. Let $\ell_s(n) \leq O(s(n) \log s(n))$ denote the number of bits needed to encode a function $f \in$ SIZE$[s(n)]$. On input $n \in \mathbb{N}$, $\varepsilon > 0$, and a set $S = (\langle x^{(1)}, b^{(1)}\rangle, \ldots, \langle x^{(m)}, b^{(m)}\rangle)$ of samples, R operates as follows.

1. Compute $t := \max\{p_0(n \cdot m^2), p_1(T(n), m)\}$, where p_0 is the polynomial from Lemma 26 and p_1 is the polynomial from Lemma 60. Also compute $t' := t'(t)$, where t' is some polynomial specified later (in Claim 62).

2. Compute

$$\beta := \mathrm{Approx}_\tau\text{-pK}(X, 1^t), \text{ and } \beta' := \mathrm{Approx}_\tau\text{-pK}(X \circ b, 1^{t'}),$$

where $X := x^{(1)} \circ \cdots \circ x^{(m)}$, $b := b^{(1)} \circ \cdots \circ b^{(m)}$, and Approx_τ-pK is the randomized algorithm from Lemma 28.

3. Output "correlative" if $\beta' - \beta \leq \theta$, where $\theta := m(1 - \varepsilon^2/8) + \ell_s(n) + n + \log \tau(t)$, and output "random" otherwise.

It is easy to verify that the running time of R is $\mathrm{poly}(n, m, T(n))$. Next, we argue its correctness.

Soundness. Suppose we are in the "correlative" case. That is, the labelled examples in S are sampled i.i.d from some distribution D' on $\{0,1\}^n \times \{0,1\}$, whose marginal on $\{0,1\}^n$ is given by some $D \in \mathrm{Samp}[T(n)]/a(n)$, and there exists $f \in \mathrm{SIZE}[s(n)]$ such that

$$\Pr_{\langle x^{(i)}, b^{(i)} \rangle \sim D'}[b^{(i)} = f(x^{(i)})] \geq \frac{1}{2} + \frac{\varepsilon}{2}.$$

In this case, by a standard concentration bound, the probability over $S \sim (D')^m$ that

$$|\{i \in [m] \mid b_i = f(x^{(i)})\}| < (1/2 + \varepsilon/4) \cdot m$$

is at most $\exp(-2m(\varepsilon/4)^2) \leq o(1)$, where the last inequality relies on our choice of m. Now observe the following.

Claim 62. There exists a polynomial t' such that for any $b \in \{0,1\}^m$ satisfying

$$|\{i \in [m] \mid b_i = f(x^{(i)})\}| \geq (1/2 + \varepsilon/4) \cdot m,$$

we have

$$\mathrm{pK}^{t'(t)}(X \circ b) \leq \mathrm{pK}^{\tau(t)}(X) + \ell_s(n) + (1 - \varepsilon^2/8) \cdot m.$$

Proof of Claim 62. Note that given X, we can compute $f(x^{(1)}), \ldots, f(x^{(m)})$ in time $\mathrm{poly}(m \cdot \ell_s(n))$ using the encoding of f, which is of $\ell_s(n)$ bits. Note that b and $f(x^{(1)}), \ldots, f(x^{(m)})$ disagree on at most $(1/2 - \varepsilon/4) \cdot m$ coordinates. Then to recover b, we can define a string $e \in \{0,1\}^m$ such that $e_i = 1$ iff $f(x^{(i)}) \neq b_i$. Note that e has hamming weight at most $(1/2 - \varepsilon/4) \cdot m$. Using the inequality

$$\sum_{i=0}^{k} \binom{m}{i} \leq 2^{H_2(k/m) \cdot m},$$

where H_2 is the binary entropy function, e can be encoded with $H_2(1/2 + \varepsilon/4) \cdot m$ bits, by lexicographic indexing among binary strings of bounded hamming weight. Using the Taylor series of H_2 in the neighborhood of $1/2$, for $\delta := \varepsilon/4$, we have

$$H_2(1/2 + \delta) = 1 - \frac{1}{2 \ln 2} \sum_{i=1}^{\infty} \frac{(2\delta)^{2i}}{i(2i-1)} \leq 1 - \frac{2}{\ln 2}\delta^2 \leq 1 - 2\delta^2.$$

Moreover, using such an encoding, it is not hard to see that e can be reconstructed in time $\mathrm{poly}(n, m)$. Therefore, given X, we can compute b in time $\mathrm{poly}(n, m)$ using strings of length $\ell_s(n)$ (which encodes f) and $(1 - \varepsilon^2/8) \cdot m$ (which encodes e). In other words, if X has probabilistic $\tau(t)$-time-bounded Kolmogorov complexity k, then

$$\mathrm{pK}^{t'(t)}(X \circ b) \leq k + \ell_s(n) + (1 - \varepsilon^2/8) \cdot m,$$

for some $t'(t) := \mathrm{poly}(\tau(t))$. This completes the proof of the claim. ◁

Now, suppose in Step 2 of R, β and β' output by the algorithm Approx_τ-pK are good approximations as given in Lemma 28. Note that this happen with high probability. Then by a union bound, with probability at least $2/3$, over the samples $S \sim (D')^m$ and the internal randomness of R, we have

$$\begin{aligned}
\beta' &\leq \mathrm{pK}^{t'}(X \circ b) && \beta' \text{ is a good approximation} \\
&\leq \mathrm{pK}^{\tau(t)}(X) + \ell_s(n) + (1 - \varepsilon^2/8) \cdot m && \text{Claim62} \\
&\leq \beta + \log \tau(t) + \ell_s(n) + (1 - \varepsilon^2/8) \cdot m && \beta \text{ is a good approximation,}
\end{aligned}$$

which implies

$$\beta' - \beta \leq (1 - \varepsilon^2/8) \cdot m + \ell_s(n) + n + \log \tau(t) = \theta,$$

and R will output "correlative".

Completeness. Suppose we are in the "random" case. That is, b is sampled from U_m. Assuming DistNP $\subseteq$ AvgBPP, by combining symmetry of information (Lemma 26) and probabilistic incompressibility (Lemma 20), for any $X \in \{0,1\}^{nm}$ and some polynomial p_w, we have

$$\Pr_{b \sim U_m}[\mathrm{pK}^{\tau(t')}(X \circ b) \geq \mathrm{pK}^{p_w(\tau(t'))}(X) + (m - 10) - \log p_w(\tau(t'))] \geq 1 - 1/8. \quad (40)$$

Let us define $p_\tau(t) := p_w(\tau(t'(t)))$. By Markov, we have

$$\begin{aligned}
&\Pr_{X \sim D^m}[\mathrm{pK}^t(X) - \mathrm{pK}^{p_\tau(t)}(X) > 8 \cdot \mathrm{psd}_{D,n}^{t,p_\tau(t)}(m)] \\
&\leq \frac{\mathbb{E}_{X \sim D^m}[\mathrm{pK}^t(X) - \mathrm{pK}^{p_\tau(t)}(X)]}{8 \cdot \mathrm{psd}_{D,n}^{t,p_\tau(t)}(m)} = 1/8. \quad (41)
\end{aligned}$$

Again, let us assume that in Step 2 of R, β and β' output by the algorithm Approx_τ-pK are good approximations. Then with with probability at least $1 - (1/8 + 1/8 + o(1)) \geq 2/3$ over $b \sim U_m$, $X \sim D^m$ and the internal randomness of R, we have

$$\begin{aligned}
\beta' &\geq \mathrm{pK}^{\tau(t')}(X \circ b) - \log \tau(t') && \beta' \text{ is a good approximation} \\
&\geq \mathrm{pK}^{p_w(\tau(t'))}(X) + m - \log(p_w(\tau(t')) \cdot \tau(t')) && \text{Equation40} \\
&= \mathrm{pK}^t(X) - (\mathrm{pK}^t(X) - \mathrm{pK}^{p_\tau(t)}(X)) + m - 10 - \log(p_\tau(t) \cdot \tau(t')) \\
&\geq \beta - 8 \cdot \mathrm{psd}_{D,n}^{t,p_\tau(t)}(m) + m - 10 - \log(p_\tau(t) \cdot \tau(t')) && \text{Equation41,}
\end{aligned}$$

which implies

$$\beta' - \beta \geq m - 8 \cdot \mathrm{psd}_{D,n}^{t,p_\tau(t)}(m) - \log(p_\tau(t) \cdot \tau(t')) - 10.$$

Now we want the above to be greater than θ. We have

$$\begin{aligned}(\beta' - \beta) - \theta &\geq (m - 8 \cdot \mathrm{psd}_{D,n}^{t,p_\tau(t)}(m) - \log(p_\tau(t) \cdot \tau(t')) - 10) \\ &\quad - (m + \ell_s(n) + n + \log \tau(t) - m\varepsilon^2/8) \\ &= m\varepsilon^2/8 - (n + \ell_s(n) + 8 \cdot \mathrm{psd}_{D,n}^{t,p_\tau(t)}(m) \\ &\quad + \log(p_\tau(t) \cdot \tau(t') \cdot \tau(t)) + 10) \\ &\geq m\varepsilon^2/8 - [n + \ell_s(n) + c \cdot (\log m + \log T(n) + a(n)) \\ &\quad + \log p_2(n, m, T(n)) + 10], \qquad \text{(Lemma60)}\end{aligned}$$

where $c > 0$ is some constant and p_2 is some polynomial (that depends on τ, p_0, p_1 and p_w). From here, it can be easily seen that the above is greater than 0 by our choice of m, provided that n is sufficiently large. Hence R will output "random". This completes the proof of the theorem. ◁

5.3 Concluding the proof

We are now ready to prove Theorem 56.

Theorem 63 (Reminder of Theorem 56). If DistNP $\subseteq$ AvgBPP, then for any time constructible functions $s, T, a : \mathbb{N} \to \mathbb{N}$, and $\varepsilon \in [0, 1]$, SIZE$[s(n)]$ is agnostic learnable on Samp$[T(n)]/a(n)$ in time poly$(n, \varepsilon^{-1}, s(n), T(n), a(n))$ with sample complexity

$$\left(\frac{(n + s(n) + a(n) + \log T(n))^3}{\varepsilon^8}\right)^{1+\zeta},$$

where $\zeta > 0$ is any arbitrary small constant.

Proof. The theorem follows by combining Theorem 61 and Theorem 58. ◁

References

[1] Luis Filipe Coelho Antunes and Lance Fortnow. Worst-case running times for average-case algorithms. In Proceedings of the 24th Annual IEEE Conference on Computational Complexity, CCC 2009, Paris, France, 15-18 July 2009, pages 298–303. IEEE Computer Society, 2009. doi:10.1109/CCC.2009.12.

[2] Shai Ben-David, Benny Chor, Oded Goldreich, and Michael Luby. On the theory of average case complexity. J. Comput. Syst. Sci., 44(2):193–219, 1992. doi:10.1016/0022-0000(92)90019-F.

[3] Andrej Bogdanov and Luca Trevisan. Average-case complexity. Found. Trends Theor. Comput. Sci., 2(1), 2006. doi:10.1561/0400000004.

[4] Andrej Bogdanov and Luca Trevisan. On worst-case to average-case reductions for NP problems. SIAM J. Comput., 36(4):1119–1159, 2006. doi:10.1137/S0097539705446974.

[5] Harry Buhrman, Lance Fortnow, and Aduri Pavan. Some results on derandomization. Theory Comput. Syst., 38(2):211–227, 2005. doi:10.1007/s00224-004-1194-y.

[6] Lijie Chen, Shuichi Hirahara, and Neekon Vafa. Average-case hardness of NP and PH from worst-case fine-grained assumptions. In Mark Braverman, editor, 13th Innovations in Theoretical Computer Science Conference, ITCS 2022, January 31 - February 3, 2022, Berkeley, CA, USA, volume 215 of LIPIcs, pages 45:1–45:16. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPIcs.ITCS.2022.45.

[7] Halley Goldberg and Valentine Kabanets. A simpler proof of the worst-case to average-case reduction for polynomial hierarchy via symmetry of information. Electron. Colloquium Comput. Complex., 7:1–14, 2022. URL: https://eccc.weizmann.ac.il/report/2022/007/.

[8] Oded Goldreich. The Foundations of Cryptography - Volume 1: Basic Techniques. Cambridge University Press, 2001. doi:10.1017/CBO9780511546891.

[9] Oded Goldreich and Leonid A. Levin. A hard-core predicate for all one-way functions. In Proceedings of the Twenty-First Annual ACM Symposium on Theory of Computing, STOC '89, pages 25–32, New York, NY, USA, 1989. Association for Computing Machinery. doi:10.1145/73007.73010.

[10] Shuichi Hirahara. Characterizing average-case complexity of PH by worst-case meta-complexity. In Sandy Irani, editor, 61st IEEE Annual Symposium on Foundations of Computer Science, FOCS 2020, Durham, NC, USA, November 16-19, 2020, pages 50–60. IEEE, 2020. doi:10.1109/FOCS46700.2020.00014.

[11] Shuichi Hirahara. Average-case hardness of NP from exponential worst-case hardness assump- tions. In Samir Khuller and Virginia Vassilevska Williams, editors, STOC '21: 53rd Annual ACM SIGACT Symposium on Theory of Computing, Virtual Event, Italy, June 21-25, 2021, pages 292–302. ACM, 2021. doi:10.1145/3406325.3451065.

[12] Shuichi Hirahara. Symmetry of information in heuristica. Manuscript, 2021.

[13] Shuichi Hirahara and Mikito Nanashima. On worst-case learning in relativized heuristica. In Symposium on Foundations of Computer Science (FOCS), 2021.

[14] Shuichi Hirahara and Rahul Santhanam. Errorless versus error-prone average-case complexity. In Mark Braverman, editor, 13th Innovations in Theoretical Computer Science Conference, ITCS 2022, January 31 - February 3, 2022, Berkeley, CA, USA, volume 215 of LIPIcs, pages 84:1–84:23. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPIcs.ITCS.2022.84.

[15] Shuichi Hirahara and Rahul Santhanam. Excluding PH pessiland. In Mark Braverman, editor, 13th Innovations in Theoretical Computer Science Conference, ITCS 2022, January 31 - February 3, 2022, Berkeley, CA, USA,

volume 215 of LIPIcs, pages 85:1–85:25. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPIcs.ITCS.2022.85.

[16] Russell Impagliazzo. A personal view of average-case complexity. In Proceedings of the Tenth Annual Structure in Complexity Theory Conference, Minneapolis, Minnesota, USA, June 19-22, 1995, pages 134–147, 1995. doi:10.1109/SCT.1995.514853.

[17] Russell Impagliazzo. Relativized separations of worst-case and average-case complexities for NP. In Proceedings of the 26th Annual IEEE Conference on Computational Complexity, CCC 2011, San Jose, California, USA, June 8-10, 2011, pages 104–114, 2011. doi:10.1109/CCC.2011.34.

[18] Russell Impagliazzo and Leonid A. Levin. No better ways to generate hard NP instances than picking uniformly at random. In 31st Annual Symposium on Foundations of Computer Science, St. Louis, Missouri, USA, October 22-24, 1990, Volume II, pages 812–821. IEEE Computer Society, 1990. doi:10.1109/FSCS.1990.89604.

[19] Russell Impagliazzo and Avi Wigderson. P = BPP if E requires exponential circuits: De- randomizing the XOR lemma. In Frank Thomson Leighton and Peter W. Shor, editors, Proceedings of the Twenty-Ninth Annual ACM Symposium on the Theory of Computing, El Paso, Texas, USA, May 4-6, 1997, pages 220–229. ACM, 1997. doi:10.1145/258533.258590.

[20] Michael J. Kearns, Robert E. Schapire, and Linda Sellie. Toward efficient agnostic learning. Mach. Learn., 17(2-3):115–141, 1994. doi:10.1007/BF00993468.

[21] Pravesh K. Kothari and Roi Livni. Improper learning by refuting. In Anna R. Karlin, editor, 9th Innovations in Theoretical Computer Science Conference, ITCS 2018, January 11-14, 2018, Cambridge, MA, USA, volume 94 of LIPIcs, pages 55:1–55:10. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2018. doi:10.4230/LIPIcs.ITCS.2018.55.

[22] Clemens Lautemann. BPP and the polynomial hierarchy. Inf. Process. Lett., 17(4):215–217, 1983. doi:10.1016/0020-0190(83)90044-3.

[23] Troy Lee. Kolmogorov complexity and formula lower bounds. PhD thesis, University of Amsterdam, 2006.

[24] Leonid A. Levin. Laws of information conservation (nongrowth) and aspects of the foundation of probability theory. Problemy Peredachi Informatsii, 10(3):30–35, 1974.

[25] Leonid A. Levin. Average case complete problems. SIAM J. Comput., 15(1):285–286, 1986. doi:10.1137/0215020.

[26] Ming Li and Paul M. B. Vitányi. An introduction to Kolmogorov complexity and its applications. Springer, 2019.

[27] Ming Li and Paul M. B. Vitányi. An Introduction to Kolmogorov Complexity and Its Applications, 4th Edition. Texts in Computer Science. Springer, 2019. doi:10.1007/978-3-030-11298-1.

[28] Yanyi Liu and Rafael Pass. On one-way functions and kolmogorov complexity. In 61st IEEE Annual Symposium on Foundations of Computer Science, FOCS 2020, Durham, NC, USA, November 16-19, 2020, pages 1243–1254, 2020. doi:10.1109/FOCS46700.2020.00118.

[29] Yanyi Liu and Rafael Pass. On one-way functions from np-complete problems. Electron. Colloquium Comput. Complex., page 59, 2021. URL: https://eccc.weizmann.ac.il/report/2021/059.

[30] Yanyi Liu and Rafael Pass. On the possibility of basing cryptography on EXP $\neq$ BPP. In Advances in Cryptology - CRYPTO 2021 - 41st Annual International Cryptology Conference, CRYPTO 2021, Virtual Event, August 16-20, 2021, Proceedings, Part I, pages 11–40, 2021. doi:10.1007/978-3-030-84242-0_2.

[31] Luc Longpré and Osamu Watanabe. On symmetry of information and polynomial time invertibility. Inf. Comput., 121(1):14–22, 1995. doi:10.1006/inco.1995.1120.

[32] Zhenjian Lu and Igor C. Oliveira. An efficient coding theorem via probabilistic representations and its applications. In Nikhil Bansal, Emanuela Merelli, and James Worrell, editors, 48th International Colloquium on Automata, Languages, and Programming, ICALP 2021, July 12-16, 2021, Glasgow, Scotland (Virtual Conference), volume 198 of LIPIcs, pages 94:1–94:20. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2021. doi:10.4230/LIPIcs.ICALP.2021.94.

[33] Zhenjian Lu, Igor C. Oliveira, and Rahul Santhanam. Pseudodeterministic algorithms and the structure of probabilistic time. In Samir Khuller and Virginia Vassilevska Williams, editors, STOC '21: 53rd Annual ACM SIGACT Symposium on Theory of Computing, Virtual Event, Italy, June 21-25, 2021, pages 303–316. ACM, 2021. doi:10.1145/3406325.3451085.

[34] Zhenjian Lu, Igor C. Oliveira, and Marius Zimand. Optimal coding theorems in time-bounded kolmogorov complexity. In 49th International Colloquium on Automata, Languages and Programming, 2022.

[35] Peter Bro Miltersen, N. V. Vinodchandran, and Osamu Watanabe. Superpolynomial versus half-exponential circuit size in the exponential hierarchy. In Takao Asano, Hiroshi Imai, D. T. Lee, Shin-Ichi Nakano, and Takeshi Tokuyama, editors, Computing and Combinatorics, 5th Annual International Conference, COCOON '99, Tokyo, Japan, July 26-28, 1999, Proceedings, volume 1627 of Lecture Notes in Computer Science, pages 210–220. Springer, 1999. doi:10.1007/3-540-48686-0_21.

[36] Ashish V. Naik, Kenneth W. Regan, and D. Sivakumar. On quasilinear-time complexity theory. Theor. Comput. Sci., 148(2):325–349, 1995. doi:10.1016/0304-3975(95)00031-Q.

[37] Noam Nisan and Avi Wigderson. Hardness vs randomness. J. Comput. Syst. Sci., 49(2):149–167, 1994. doi:10.1016/S0022-0000(05)80043-1.

[38] Igor C. Oliveira. Randomness and intractability in kolmogorov complexity. In Christel Baier, Ioannis Chatzigiannakis, Paola Flocchini, and Stefano Leonardi, editors, 46th International Colloquium on Automata, Languages, and Programming, ICALP 2019, July 9-12, 2019, Patras, Greece, volume 132 of LIPIcs, pages 32:1–32:14. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2019. doi:10.4230/LIPIcs.ICALP.2019.32.

[39] Igor C. Oliveira and Rahul Santhanam. Pseudodeterministic constructions in subexponential time. In Hamed Hatami, Pierre McKenzie, and Valerie King, editors, Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2017, Montreal, QC, Canada, June 19-23, 2017, pages 665–677. ACM, 2017. doi:10.1145/3055399.3055500.

[40] Hanlin Ren and Rahul Santhanam. Hardness of KT characterizes parallel cryptography. In 36th Computational Complexity Conference, CCC 2021, July 20-23, 2021, Toronto, Ontario, Canada (Virtual Conference), pages 35:1–35:58, 2021. doi:10.4230/LIPIcs.CCC.2021.35.

[41] Shai Shalev-Shwartz and Shai Ben-David. Understanding Machine Learning - From Theory to Algorithms. Cambridge University Press, 2014. URL: http://www.cambridge.org/de/academic/subjects/computer-science/pattern-recognition-and-machine-learning/ understanding-machine-learning-theory-algorithms.

[42] Ronen Shaltiel and Christopher Umans. Simple extractors for all min-entropies and a new pseudorandom generator. J. ACM, 52(2):172–216, 2005. doi:10.1145/1059513.1059516.

[43] Luca Trevisan and Salil P. Vadhan. Pseudorandomness and average-case complexity via uniform reductions. Computational Complexity, 16(4):331–364, 2007.

[44] Salil P. Vadhan. On learning vs. refutation. In Satyen Kale and Ohad Shamir, editors, Proceed- ings of the 30th Conference on Learning Theory, COLT 2017, Amsterdam, The Netherlands, 7-10 July 2017, volume 65 of Proceedings of Machine Learning Research, pages 1835–1848. PMLR, 2017. URL: http://proceedings.mlr.press/v65/vadhan17a.html.

[45] Emanuele Viola. On constructing parallel pseudorandom generators from one-way functions. In 20th Annual IEEE Conference on Computational Complexity (CCC 2005), 11-15 June 2005, San Jose, CA, USA, pages 183–197, 2005. doi:10.1109/CCC.2005.16.